TK-Shield

Defensive Bio-Piracy Monitoring for Traditional Knowledge

A keyless, offline-first AI system for the defensive protection of documented traditional knowledge — aligned with the WIPO IGC and the 2024 Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge.

Ansul Suryawanshi · Project brief · June 2026

The problem. Communities have practised and documented traditional medicinal and agricultural knowledge for centuries, yet patents are repeatedly granted over that same knowledge. The landmark turmeric, neem, and basmati patents were each eventually revoked — but only after costly, multi-year legal challenges that assembled prior-art evidence by hand. TK-Shield automates the defensive workflow: given a documented practice, it finds the patents that may claim it, scores bio-piracy risk, gathers citable prior-art evidence, and drafts an assessment and patent opposition — running entirely on free, open data and a local model so any community can use it.

Alignment with international IP policy

- **WIPO IGC** — the Intergovernmental Committee on IP, Genetic Resources, Traditional Knowledge and Folklore, whose central goal is preventing the erroneous granting of patents over TK.
- **WIPO GRATK Treaty (2024)** — introduces a disclosure-of-origin requirement for patents based on genetic resources and associated TK; TK-Shield assembles exactly that origin evidence.
- **Nagoya Protocol / CBD** — benefit-sharing depends on identifying the holder community; TK-Shield surfaces documented communities & peoples as a first-class attribution dimension.
- **TKDL** — India's Traditional Knowledge Digital Library is the proven model; TK-Shield mirrors its defensive purpose with open data and open models.

How the system works

Documented TK practice	Hybrid search (semantic+BM25)	5-factor risk score	Prior-art (PubMed/Wikidata /GBIF)	RAG report (local LLM)	Dashboard (3 personas)

A documented practice flows through a hybrid search engine that fuses dense semantic retrieval with lexical BM25 via Reciprocal Rank Fusion — recovering folk, multilingual, and scientific synonyms a single method would miss. A transparent five-factor model (similarity, temporal proximity, geographic overlap, assignee profile, and patent classification) produces a 0–100 risk score and a MINIMAL→CRITICAL band. Keyless prior-art enrichment fans out to PubMed, Wikidata, and GBIF, returning one deduplicated, citation-tagged evidence bundle. Finally a local LLM (Ollama) writes a citation-backed assessment and a draft opposition; with no model present it falls back to a deterministic template, keeping figures and citations exact.

Three personas, one platform. A **Defender** (community/NGO) registers a practice and obtains a risk report and opposition draft; an **Examiner** (patent office) pastes a patent and receives a novelty verdict against the TK registry; a **Researcher** explores corpus and registry analytics. The system indexes **16,371** real US patents and **2,030** documented TK practices, with zero API keys.

Evaluation — re-identifying the landmark cases

Each of the three landmark bio-piracy cases was submitted as an **independently-worded** TK description sharing no wording with the patent, then run through the full pipeline over the 16,371-patent corpus (retrieval depth k=10). Retrieval therefore reflects genuine semantic and lexical matching, not string overlap.

100%	67%	0.833	100%
Precision@5	Precision@1	Mean Recip. Rank	Flagged HIGH/CRITICAL

TK practice	Patent (claimant)	Rank	Sim.	Risk	Historical outcome
Turmeric	US5401504A University of Mississippi Medical Center (US)	#1	0.823	**CRITICAL (85)**	Revoked by the USPTO in 1997 on documented Indian prior art.
Neem	EP0436257B1 W.R. Grace & Co. / USDA (US)	#1	0.787	**CRITICAL (83)**	Revoked by the EPO in 2000/2005 on documented Indian prior art.
Basmati aromatic rice of the Indian subcontinent	US5663484A RiceTec Inc. (US)	#2	0.606	**CRITICAL (83)**	Most claims withdrawn/struck at the USPTO in 2001-2002.

From folk-worded inputs alone, TK-Shield re-identifies all three historically-revoked patents — every one retrieved within the top 10 (Precision@5 100%), 67% as the single closest match — and scores 100% in the HIGH/CRITICAL band. The result is reproducible via `python -m src.evaluation.landmark_eval`.

Engineering principles

- **Free & keyless first** — the entire pipeline runs with zero API keys on public-domain / open data (PatentsView, Dr. Duke CC0, Wikidata) and a local model; no paid or gated services.
- **Graceful degradation** — no external source or the LLM can crash the pipeline; reports note any skipped source, and everything works offline.
- **Citations, not hand-waving** — every claim carries a stable reference (PMID / Wikidata QID / GBIF key / patent number).
- **Security by construction** — server/LLM/user text is never rendered as raw HTML, external links are scheme-validated, and request inputs are bounded.
- **Tested** — 48 network-free backend tests and 21 frontend tests, including XSS-safety and the landmark-evaluation regressions above.

Impact and future work

TK-Shield lowers the cost of defensive TK protection from a specialist legal exercise to a tool a community can run on a laptop, and provides patent examiners a fast prior-art check at the point of decision — directly supporting the disclosure-of-origin objective of the 2024 WIPO treaty. Current scope is US patent metadata, English-language analysis, and a small local model; natural extensions are full-text and non-US patent coverage, multilingual TK ingestion at scale, continuous monitoring with alerting, and a multi-tenant deployment for institutional use. The architecture isolates data access behind clean interfaces so these extensions are localized changes.

Data: PatentsView (USPTO, public domain); Dr. Duke's Phytochemical & Ethnobotanical Databases (USDA, CC0); Wikidata (CC0); enrichment via PubMed, Wikidata, and GBIF. TK-Shield is a defensive research tool and does not constitute legal advice.